

# Candidate Project Submission

## Plug-and-Play Sensing & Control Slice for a Growing Rack

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# 1 Component Selection Matrix

## 1.1 Design

The proposed sensing and control slice monitors two environmental parameters—air temperature and relative humidity—and controls a single exhaust fan. Home Assistant combines both measurements to calculate Vapor Pressure Deficit (VPD), which serves as the control variable for rack ventilation.

|                          |                                             |
|--------------------------|---------------------------------------------|
| Environmental Parameters | Air Temperature, Relative Humidity          |
| Actuator                 | Exhaust Fan                                 |
| Control Variable         | Vapor Pressure Deficit (VPD)                |
| Home Assistant Role      | Sensor fusion, automation, actuator control |

Table 1: Selected sensing and control architecture.

## 1.2 Temperature and Humidity Sensor Evaluation

| Product           | Vendor       | Price   | Link         |
|-------------------|--------------|---------|--------------|
| Adafruit SHT45    | Adafruit     | \$12.50 | Adafruit     |
| Adafruit SHT31-D  | Adafruit     | \$13.95 | Adafruit     |
| Seeed Grove SHT40 | Seeed Studio | \$5.50  | Seeed Studio |

Table 2: Temperature and humidity sensor candidates.

| Product           | HA | Data | Reliability | Cost | Availability |
|-------------------|----|------|-------------|------|--------------|
| Adafruit SHT45    | 5  | 5    | 5           | 4    | 5            |
| Adafruit SHT31-D  | 5  | 4    | 5           | 4    | 5            |
| Seeed Grove SHT40 | 5  | 4    | 4           | 5    | 4            |

Table 3: Evaluation scores (5 = best).

**Selected Sensor: Adafruit SHT45** The Adafruit SHT45 was selected for the proposed design due to its superior accuracy, reliability, and straightforward ESPHome integration. The sensor includes factory calibration with ISO 17025 traceability, supports operation in condensing environments, and communicates over a standard I<sup>2</sup>C interface that is natively supported by ESPHome.

### 1.3 ESP32 Bridge Evaluation

| Product                  | Vendor    | Price | Link    |
|--------------------------|-----------|-------|---------|
| ESP32-DEVKITM-1          | Espressif | \$8   | DigiKey |
| ESP32-S3-DEVKITC-1U-N8R8 | Espressif | \$15  | DigiKey |
| ESP32-C3-DEVKITM-1-N4X   | Espressif | \$8   | DigiKey |

Table 4: ESP32 bridge candidates.

| Product                  | HA | Performance | Reliability | Cost | Availability |
|--------------------------|----|-------------|-------------|------|--------------|
| ESP32-DEVKITM-1          | 5  | 5           | 5           | 5    | 5            |
| ESP32-S3-DEVKITC-1U-N8R8 | 5  | 5           | 5           | 4    | 4            |
| ESP32-C3-DEVKITM-1-N4X   | 5  | 4           | 4           | 5    | 5            |

Table 5: ESP32 bridge evaluation scores (5 = best).

**Selected Bridge: ESP32-DEVKITM-1** The ESP32-DEVKITM-1 was selected because it is a mature and widely adopted platform with excellent ESPHome support. It provides more than sufficient processing capability for an I<sup>2</sup>C sensor bridge while maintaining a lower cost than the ESP32-S3. Compared to the ESP32-C3, it offers greater flexibility for future expansion while remaining broadly available from distributors such as DigiKey and Mouser.

### 1.4 Smart Plug Evaluation

| Product             | Vendor  | Price | Link    |
|---------------------|---------|-------|---------|
| Shelly Plug US Gen4 | Shelly  | \$25  | Shelly  |
| Eve Energy          | Eve     | \$40  | Eve     |
| TP-Link Tapo TP15   | TP-Link | \$20  | TP-Link |

Table 6: Smart plug candidates.

| Product             | HA | Telemetry | Reliability | Cost | Availability |
|---------------------|----|-----------|-------------|------|--------------|
| Shelly Plug US Gen4 | 5  | 5         | 5           | 4    | 5            |
| Eve Energy          | 5  | 5         | 5           | 3    | 4            |
| TP-Link Tapo TP15   | 5  | 4         | 4           | 5    | 5            |

Table 7: Smart plug evaluation scores (5 = best).

**Selected Smart Plug: Shelly Plug US Gen4** The Shelly Plug US Gen4 aligns closely with Luce Labs’ preference for open protocols by supporting Matter while operating entirely on the local network. It integrates directly into Home Assistant, includes power monitoring for basic actuator health checks, and can be replaced by another Matter-compatible smart plug without modifying the automation logic.

## 1.5 Exhaust Fan Evaluation

| Product                  | Vendor      | Price | Link        |
|--------------------------|-------------|-------|-------------|
| AC Infinity AXIAL 1238   | AC Infinity | \$39  | AC Infinity |
| AC Infinity CLOUDLINE S6 | AC Infinity | \$100 | AC Infinity |
| VIVOSUN AeroZesh G6      | VIVOSUN     | \$104 | VIVOSUN     |

Table 8: Exhaust fan candidates.

| Product                  | Control | Reliability | Noise | Cost | Availability |
|--------------------------|---------|-------------|-------|------|--------------|
| AC Infinity AXIAL 1238   | 5       | 5           | 4     | 5    | 5            |
| AC Infinity CLOUDLINE S6 | 5       | 5           | 5     | 2    | 5            |
| VIVOSUN AeroZesh G6      | 5       | 4           | 4     | 2    | 5            |

Table 9: Exhaust fan evaluation scores (5 = best).

**Selected Fan: AC Infinity AXIAL 1238** The AXIAL 1238 was selected because it provides reliable continuous ventilation while remaining inexpensive and simple to control through a Matter-compatible smart plug. Keeping the fan as a basic AC load ensures that all control logic remains centralized within Home Assistant rather than embedded in proprietary vendor ecosystems.

## 1.6 Weakest Buy vs. Build Decision

The weakest buy in the proposed architecture is the ESP32-based sensing node. Although the combination of an ESP32 and SHT45 provides excellent performance and integrates cleanly into Home Assistant through ESPHome, it introduces an additional microcontroller, firmware maintenance, and WiFi dependency.

For a single rack or small deployment, this approach aligns well with Luce Labs’ “buy before build” philosophy. However, for deployments consisting of dozens or hundreds of racks, a custom sensing node would become attractive. A purpose-built PCB integrating the temperature and humidity sensor, Matter connectivity, watchdog protection, industrial connectors, and standardized mounting would reduce cost, simplify installation, and

improve long-term maintainability while continuing to expose standard Home Assistant entities.

## 2 Integration and Control Design

### 2.1 System Architecture

Figure 1 shows the sensing and control slice for one growing rack. The Adafruit SHT45 measures temperature and relative humidity through I<sup>2</sup>C and connects to an ESP32-DEVKITM-1 running ESPHome. The ESP32 publishes sensor data to Home Assistant over WiFi. Home Assistant calculates VPD, executes automation logic, and controls the AC Infinity AXIAL 1238 exhaust fan through a Matter command sent to the Shelly Plug US Gen4. US Gen4.

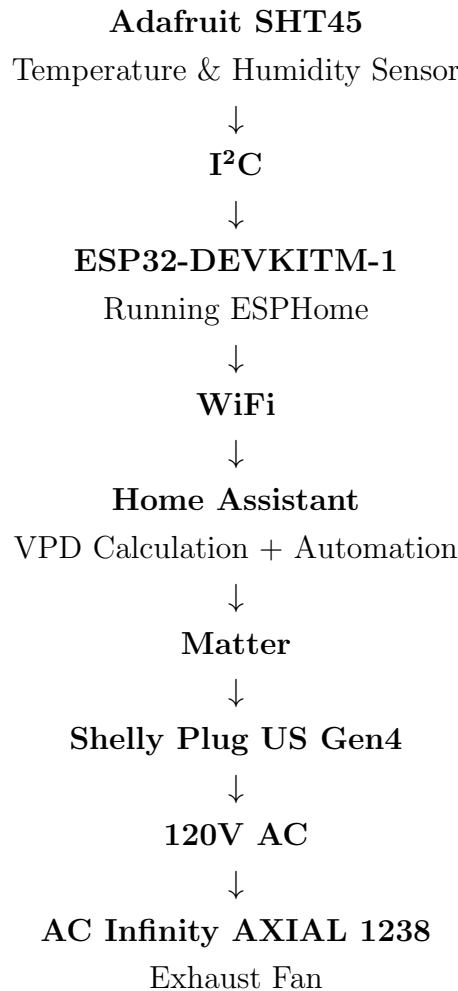


Figure 1: Proposed sensing and control architecture.

## 2.2 Home Assistant Integration

### Temperature and Humidity Sensor

The Adafruit SHT45 communicates with the ESP32 using the I<sup>2</sup>C interface. ESPHome periodically polls the sensor and publishes the following entities to Home Assistant:

- `sensor.rack_temperature`
- `sensor.rack_humidity`

These entities are automatically discovered through the native ESPHome integration and behave identically to native Home Assistant sensors.

### ESP32 Bridge

The ESP32-DEVKITM-1 runs ESPHome firmware and connects to Home Assistant through native integration. ESPHome provides automatic entity discovery and OTA firmware updates without requiring MQTT middleware.

### Smart Plug

The Shelly Plug US Gen4 joins Home Assistant using the Matter integration and exposes the following entities:

- `switch.rack_fan`
- `sensor.rack_fan_power`
- `sensor.rack_fan_energy`

The smart plug acts solely as the actuator interface. All environmental decision making remains centralized within Home Assistant.

## 2.3 Non-Native Sensor Bridge

The Adafruit SHT45 is not a native Matter device, so the ESP32-DEVKITM-1 acts as the hardware and firmware bridge. ESPHome reads the sensor through I<sup>2</sup>C and publishes the measurements as standard Home Assistant entities.

Home Assistant interacts only with the exposed entities, making the sensing node functionally identical to a native HA device. The sensor hardware can therefore be replaced without modifying dashboards or automation logic.

## Bridge Selection

| Option         | Reason Not Selected                                                                                         |
|----------------|-------------------------------------------------------------------------------------------------------------|
| MQTT Discovery | Requires an MQTT broker and additional configuration overhead.                                              |
| Matter Bridge  | Adds unnecessary complexity for a simple I <sup>2</sup> C sensor node.                                      |
| ESPHome        | Native Home Assistant support, automatic entity discovery, OTA firmware updates, and minimal configuration. |

Table 10: Comparison of sensor bridging approaches.

## 2.4 Correlation Logic

Home Assistant combines temperature and relative humidity measurements to calculate Vapor Pressure Deficit (VPD). VPD provides a more meaningful indicator of plant transpiration conditions than either parameter alone.

The exhaust fan is controlled according to the following thresholds.

| Condition                    | Action               |
|------------------------------|----------------------|
| VPD > 1.2 kPa for 5 minutes  | Turn exhaust fan ON  |
| VPD < 1.0 kPa for 10 minutes | Turn exhaust fan OFF |

Table 11: VPD control thresholds.

The separation between ON and OFF thresholds introduces hysteresis, preventing rapid switching. Minimum activation times reduce short cycling and extend actuator lifetime.

## 2.5 Failure Modes

| Failure Mode                     | Detection                                                                        | Response                                                                  |
|----------------------------------|----------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Temperature sensor offline       | Entity unavailable                                                               | Turn fan ON and generate Home Assistant alert.                            |
| Humidity sensor offline          | Entity unavailable                                                               | Turn fan ON and generate Home Assistant alert.                            |
| Implausible sensor reading       | Temperature/RH outside expected operating range or unchanged for extended period | Ignore invalid data, generate alert, and maintain fan ON as a safe state. |
| ESP32 offline                    | Device unavailable                                                               | Turn fan ON and notify operator.                                          |
| Fan commanded ON but drawing 0 W | Shelly power monitoring                                                          | Generate maintenance alert indicating possible actuator failure.          |
| Home Assistant restart           | Automation reload                                                                | Restore previous actuator state.                                          |
| WiFi network loss                | Device unavailable                                                               | Default fan to ON until communication is restored.                        |

Table 12: Failure modes and safe-state responses.

The proposed design adopts a ventilation-first safe state because excessive humidity presents a greater long-term risk to unattended plant growth than continuous airflow.

## 2.6 Bench Testing

Before deployment, the following tests would be performed:

1. Verify SHT45 measurements against a reference sensor.
2. Confirm ESPHome discovery and HA entity creation.
3. Validate VPD calculations using known temperature/RH values.
4. Test fan activation, hysteresis, and anti-short-cycling behavior.
5. Disconnect sensors and verify safe-state behavior.
6. Test WiFi loss and Home Assistant restart recovery.
7. Perform a 24-hour continuous operation test.

## 2.7 Architectural Alignment

This design aligns with Luce Labs' plug-and-play architecture by keeping Home Assistant as the central control layer and using open protocols including I<sup>2</sup>C, WiFi, ESPHome, and Matter. The ESP32 bridge converts a non-native sensor into a first-class Home Assistant device while preserving hardware flexibility and scalability.

## 3 Working Configuration

### 3.1 Home Assistant Artifacts

The Home Assistant configuration is organized into separate files for sensor integration, data processing, automation, and visualization.

- `esphome/rack-sensor.yaml`: ESPHome firmware configuration for the ESP32 bridge and SHT45 temperature/humidity sensor.
- `template_sensors.yaml`: Creates the VPD calculation entity from temperature and humidity measurements.
- `automations.yaml`: Implements the VPD-based exhaust fan control logic with hysteresis and anti-short-cycling.
- Dashboard sketch: Visual representation of the Home Assistant Lovelace dashboard layout.

### 3.2 Dashboard Sketch

The dashboard provides operators with a real-time overview of environmental conditions, actuator status, and system health.

|               |                        |  |
|---------------|------------------------|--|
| +-----+       |                        |  |
|               | Growing Rack Dashboard |  |
| +-----+       |                        |  |
| Temperature:  | 24.3 °C                |  |
| Humidity:     | 62 %                   |  |
| VPD:          | 1.08 kPa               |  |
| +-----+       |                        |  |
| Exhaust Fan:  | OFF                    |  |
| Fan Power:    | 0 W                    |  |
| Energy Usage: | 0.08 kWh               |  |
| +-----+       |                        |  |

|                                     |  |
|-------------------------------------|--|
| System Status                       |  |
| ESP32 Bridge: Online                |  |
| SHT45 Sensor: Connected             |  |
| Shelly Plug: Connected              |  |
| Home Assistant: Running             |  |
| +-----+                             |  |
| Historical Data                     |  |
| Temperature / Humidity / VPD Trends |  |
| +-----+                             |  |

## 4 AI Usage Appendix

AI tools were used to accelerate component research, Home Assistant configuration development, and document refinement.

### 4.1 Tools Used

- ChatGPT: Used for architecture brainstorming, ESPHome YAML drafting, and checking Home Assistant configuration consistency.
- Web search: Used to verify product availability, pricing, and protocol support.

### 4.2 Important Prompts

Example prompts used:

1. "Design a Home Assistant architecture using a non-native I2C sensor bridged through ESPHome."
2. "Generate a Home Assistant automation using VPD thresholds with hysteresis."
3. "Compare Matter smart plugs suitable for Home Assistant."

### 4.3 AI Verification

One issue encountered was that AI-generated suggestions occasionally referenced unsupported Home Assistant integrations or inaccurate product capabilities. These outputs were verified against vendor documentation before inclusion.

For example, product protocol support and Home Assistant compatibility were manually checked before selecting the Shelly Plug US Gen4.

## 4.4 Work Not Delegated

The final architecture decisions, product selection, tradeoffs, and failure-mode analysis were completed manually. AI was used as a productivity tool rather than replacing engineering judgment.

## 5 Time Allocation

The project was completed within the 6–8 hour time constraint.

| Task                                     | Time Spent |
|------------------------------------------|------------|
| Component research and evaluation        | 2 hours    |
| Architecture design and block diagram    | 1 hour     |
| Home Assistant and ESPHome configuration | 2 hours    |
| Failure analysis and testing plan        | 1 hour     |
| Documentation and review                 | 1 hour     |
| Total                                    | 7 hours    |

Table 13: Project time allocation.

## A Component Links

The following links correspond to the products evaluated in the component selection matrix.

### A.1 Temperature and Humidity Sensors

- Adafruit SHT45: [Adafruit SHT45 Temperature & Humidity Sensor](#)
- Adafruit SHT31-D: [Adafruit SHT31-D Temperature & Humidity Sensor](#)
- Seeed Grove SHT40: [Seeed Grove SHT40](#)

### A.2 ESP32 Development Boards

- ESP32-DEVKITM-1: [DigiKey ESP32-DEVKITM-1](#)
- ESP32-S3-DEVKITC-1U-N8R8: [DigiKey ESP32-S3-DEVKITC-1U-N8R8](#)
- ESP32-C3-DEVKITM-1-N4X: [DigiKey ESP32-C3-DEVKITM-1-N4X](#)

### A.3 Smart Plugs

- Shelly Plug US Gen4: [Shelly Plug US Gen4](#)
- Eve Energy: [Eve Energy](#)
- TP-Link Tapo TP15: [TP-Link Tapo Smart Plug](#)

### A.4 Exhaust Fans

- AC Infinity AXIAL 1238: [AC Infinity AXIAL 1238](#)
- AC Infinity CLOUDLINE S6: [AC Infinity CLOUDLINE S6](#)
- VIVOSUN AeroZesh: [VIVOSUN AeroZesh G6](#)